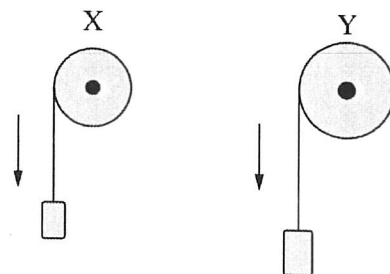
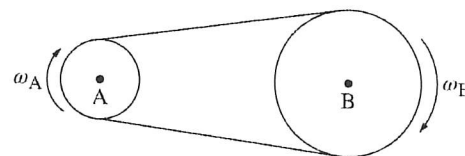


1. Pulleys X and Y are each attached to a block by a string that wraps around the pulley. Both blocks are released from rest and have the same linear acceleration  $a$ . As the blocks fall, the pulleys rotate about their centers. Pulley Y has a larger radius than Pulley X. How does the final angular velocity  $\omega_Y$  of Pulley Y compare to the final angular velocity  $\omega_X$  of Pulley X?



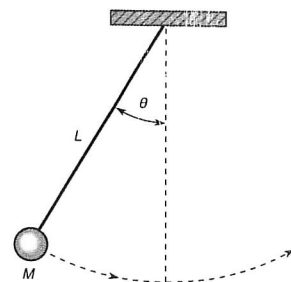
- (A)  $\omega_Y > \omega_X$   
(B)  $\omega_Y < \omega_X$   
(C)  $\omega_Y = \omega_X$   
(D) It cannot be determined without knowing the relative masses of each block.

2. Wheels A and B are connected by a moving belt and are both free to rotate about their centers, as shown. The belt does not slip on the wheels. The radius of Wheel B is three times larger than the radius of Wheel A. Wheel A has constant angular speed  $\omega_A$  and Wheel B has constant angular speed  $\omega_B$ . Which of the following correctly relates  $\omega_A$



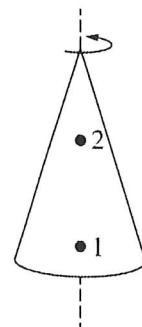
- (A)  $\omega_A = \frac{\omega_B}{3}$   
(B)  $\omega_A = \omega_B$   
(C)  $\omega_A = 3\omega_B$   
(D)  $\omega_A = 6\omega_B$

3. A pendulum is formed from a mass  $M$  connected to a string of length  $L$ . The pendulum is initially raised to form an angle  $\theta$  (measured in radians) and released. The time it takes the pendulum to swing to the opposite side and return to its initial position is measured to be  $T$  (measured in seconds). Which of the following is a correct expression for the average linear speed of the pendulum?



- (A)  $\frac{\theta}{T}$   
(B)  $\frac{\theta L}{T}$   
(C)  $\frac{2\theta L}{T}$   
(D)  $\frac{4\theta L}{T}$

4. A solid, uniform cone is spinning with constant angular speed about a vertical axis through its center of mass, as shown. A small insect is initially standing on the surface of the cone at Point 1. At a later time, the insect moved to Point 2. Which of the following statements about the motion of the insect is true?



- (A) The angular speed of the insect is greater at Point 1 than at Point 2.  
(B) The angular speed of the insect is greater at Point 2 than at Point 1.  
(C) The linear speed of the insect is greater at Point 1 than at Point 2.  
(D) The linear speed of the insect is greater at Point 2 than at Point 1.

## B Side

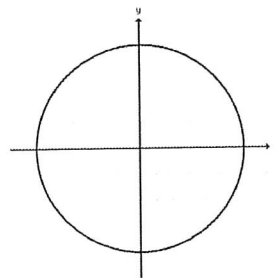
1. A disk has a radius of 2.5 m and undergoes an angular displacement of  $3\pi/4$  radians.

- Sketch this angular displacement
- Convert this measurement to degrees.

$$180 \times \frac{3}{4}\pi = 135^\circ$$

- Determine the arc length formed.

$$\text{arc} = \theta \cdot r = 135^\circ \times 2.5\text{m}$$



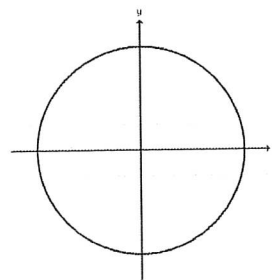
2. A woman walks in a circular path of radius 20 m. The woman undergoes an angular displacement of 2.2 rotations in the counterclockwise direction, which takes her 20 seconds to complete.

- Sketch this angular displacement
- Convert her angular displacement to radians.

$$\Rightarrow 4\pi \approx 13.816^\circ$$

- Calculate the woman's linear displacement.

$$x = \theta \cdot r = 13^\circ \times 20\text{m}$$



- Calculate the woman's angular velocity (in rad/s)

$$\omega = \frac{\Delta\theta}{t} = \frac{13^\circ}{20\text{s}}$$

- Calculate the woman's linear velocity (in m/s)

$$v = \omega \cdot r = 0.65\text{rad/s} \times 20\text{m}$$

3. A record has a small red dot on its edge. You measure that the dot travels a distance of 1.3 m (clockwise) when the record makes one full rotation, which takes 1.2 seconds.

- Determine the radius of the record

$$\omega = 6.18 \div 1.2\text{s}$$

$$r = \omega \cdot v = 5.15 \times \frac{1.3\text{m}}{1.2\text{s}}$$

- Determine the angular velocity of the record (in deg/s)

$$2\pi \div 1.2\text{s}$$

